11. How many moles of I2 are liberated when 1 mole of potassium dichromate react with potassium iodide?

a) 1 b) 2

c) 3 d) 4

12. The number of moles of acidified $KMnO_4$ required to oxidize 1 mole of ferrous oxalate(FeC_2O_4) is

a) 5 b) 3 c) 0.6 d) 1.5

13. Which one of the following statements related to lanthanons is incorrect?

a) Europium shows +2 oxidation state.

b) The basicity decreases as the ionic radius decreases from Pr to Lu.

c) All the lanthanons are much more reactive than aluminium.

d) Ce^{4+} solutions are widely used as oxidising agents in volumetric analysis.

14. Which of the following lanthanoid ions is diamagnetic?

a) Eu^{2+} b) Yb^{2+}

c) Ce^{2+} d) Sm^{2+}

15. Which of the following oxidation states is most common among the lanthanoids?

a) +4 b) +2

c) +5 d) +3

16. Assertion : Ce^{4+} is used as an oxidizing agent in volumetric analysis.
Reason: Ce^{4+} has the tendency of attaining +3 oxidation state.

a) Both assertion and reason are true and reason is the correct explanation of assertion.

b) Both assertion and reason are true but reason is not the correct explanation of assertion.

c) Assertion is true but reason is false.

d) Both assertion and reason are false.

17. The most common oxidation state of actinoids is

a) +2 b) +3

c) +4 d) +6

18. The actinoid elements which show the highest oxidation state of +7 are

a) Np, Pu ,Am b) U, Fm, Th

c) U, Th, Md d) Es, No, Lr

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